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<sol_start id=1>

1. (4) +0.5

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<sol_start id=2>

2. (2) Outer surface of the cornea.

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3. (4) Virtual, erect and enlarged.

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<sol_start id=4>

4. (1) Both A and R are true and R is the correct explanation of A.

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<sol_start id=5>

5. (3) Assertion is true but reason is false.

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<sol_start id=6>

6. The two most common defect of vision are hypermetropia and myopia. Myopia can be corrected by using a concave lens and hypermetropia can be corrected by using convex lens.

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7. (a) Water is optically denser medium compared to air. The light from the part of pencil which is immersed in water travels from water to air causing bending of light.

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(b) Given $\Rightarrow$ magnification = +2.0

$u = ?$

$v = 20 \text{ cm}$

Using $m = \frac{v}{u}$

$$2.0 = \frac{20}{u}$$

$$u = \frac{20}{2.0} = 10 \text{ cm}$$

Object distance from the lens is 10 cm.

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8. Given $\Rightarrow f = +15 \text{ cm}$

$$\Rightarrow u = -10 \text{ cm}$$

$$\text{Using } \Rightarrow \frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\Rightarrow \frac{1}{15} = \frac{1}{v} + \frac{1}{(-10)}$$

$$\frac{1}{v} = \frac{1}{15} + \frac{1}{10}$$

$$\frac{1}{v} = \frac{2+3}{30} \Rightarrow \frac{5}{30}$$

$$v = \frac{30}{5} = +6 \text{ cm}$$

The image formed behind the mirror, It is virtual and erect image.

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9. Given $\Rightarrow u = -10$

$$m = -3, v = ?, f = ?$$

$$\text{Using, } m = \frac{v}{u}$$

$$-3 = \frac{v}{-10}$$

$$+30 = v$$

Image distance from the lens is 30 cm.

$$\text{Using } \Rightarrow \frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{f} = \frac{1}{30} - \frac{1}{(-10)}$$

$$\frac{1}{f} = \frac{1+3}{30} = \frac{4}{30}$$

$$f = 7.5 \text{ cm}$$

The focal length of the lens is 7.5 cm.

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10. Myopia can be corrected by using concave lens. The lens must be placed that an object at infinity forms its image on far point.

Hence, $f = -80 \text{ cm}$

$$f = -0.8 \text{ m}$$

$$P = \frac{1}{f}$$

$$P = \frac{1}{-0.8\text{m}} \Rightarrow -1.25\text{D}$$

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11. (a) Convex lens

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(b) The image is real and inverted in nature hence $\frac{v}{u}$ (magnification) is negative.

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$$\text{(c) Using, } \frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{20} = \frac{1}{v} - \frac{1}{-21}$$

$$\frac{1}{v} = \frac{1}{20} - \frac{1}{21}$$

$$\frac{1}{v} = \frac{21-20}{420}$$

$$v = 420 \text{ cm}$$

Screen should be placed 20 cm away from the slide projector.

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12. (a) Refractive index is defined as the ratio of the speed of light in vacuum to its speed in a specific medium.

$$\text{Refractive index (n)} = \frac{\text{speed of light in vacuum}}{\text{speed of light in medium}} \Rightarrow \frac{c}{v}$$

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$$(b) \text{ Given } \Rightarrow \text{ refractive index of water } \Rightarrow n_w = \frac{4}{3}$$

$$\text{refractive index of glass } \Rightarrow n_g = \frac{3}{2}$$

$$\text{speed of light in air} = 3 \times 10^8 \text{ ms}^{-1}$$

$$\text{speed of light in glass} = 2 \times 10^8 \text{ ms}^{-1}$$

$$n_g = \frac{c}{v_g}$$

$$c = v_g \times n_g$$

$$c = \frac{2 \times 10^8 \times 3}{2}$$

$$c = 3 \times 10^8 \text{ ms}^{-1}$$

$$\text{speed of light in air } \Rightarrow 3 \times 10^8 \text{ m/s}$$

$$n_w = \frac{c}{v_w}$$

$$v_w = \frac{c}{n_w}$$

$$v_w = \frac{3 \times 10^8 \times 3}{4}$$

Speed of light in water $\Rightarrow 2.25 \times 10^8 \text{ ms}^{-1}$

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